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PHY 121 — Past Questions & Materials
by Renance

Course: PHY 121
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 2
Levels: 100

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Paper 1 of 2 — PHY121 POSSIBLE TEST Q&A BY VOTU JERRY

Level 100 • Second Semester • PastQuestion

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TESTQUESTIONS

1.
)Whatamountofcur r
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5cm, for
 to produce a magnetic
 field
 of 400mT?
 a) 762A b) 324A c) 965A d) 198A
 2.) A particle
 of charge $6.1 \mu\text{C}$ and mass $5.4 \times 10^{-18} \text{ kg}$
 is subjected to a
 uniform
 magnetic field
 of 0.1 T
 directed out of the
 page. Calculate
 the magnitude of
 the magnetic
 force
 on the particle
 if it is moving
 with a speed of
 200 m/s .
 Assume that the
 particle is moving in a
 circular orbit in
 the plane of the
 page.
 a) $0.1 \times 10^{-3} \text{ T}$ b) $2.7 \times 10^{-2} \text{ T}$ c) $4.31 \times 10^{-9} \text{ T}$ d) $3.5 \times 10^{-1} \text{ T}$
 Answer:
 The particle is moving in a
 circular orbit in the
 plane of the page.
 The magnetic force
 provides the centripetal
 force.
 The magnetic force
 is given by:
 $F = qvB \sin \theta$
 where q is the charge,
 v is the speed,
 B is the magnetic
 field, and θ is the
 angle between the
 velocity and the
 magnetic field.
 In this case, $\theta = 90^\circ$,
 so $\sin \theta = 1$.
 The centripetal force
 is given by:
 $F = \frac{mv^2}{r}$
 where m is the mass,
 v is the speed,
 and r is the radius.
 Setting the magnetic
 force equal to the
 centripetal force,
 we get:
 $q v B = \frac{m v^2}{r}$
 Solving for r , we get:
 $r = \frac{m v}{q B}$
 Substituting the
 given values, we get:
 $r = \frac{(5.4 \times 10^{-18} \text{ kg})(200 \text{ m/s})}{(6.1 \times 10^{-6} \text{ C})(0.1 \text{ T})}$
 $r = 1.77 \times 10^{-13} \text{ m}$
 The magnitude of
 the magnetic force
 is:
 $F = q v B = (6.1 \times 10^{-6} \text{ C})(200 \text{ m/s})(0.1 \text{ T})$
 $F = 1.22 \times 10^{-4} \text{ N}$

ty of $2.7 \times 10^9 \text{ m/s}$ along
 the axis of a conductor carrying
 current i
 is a current of 11 A .
 If the part
 of the wire is
 at a distance
 of 4.7 mm from the
 center of the wire. Determine
 the magnetic field
 at the point
 P. The magnetic field
 is
 a) $2.31 \times 10^{-6} \text{ T}$
 b) $7.4 \times 10^{-6} \text{ T}$
 c) $4.2 \times 10^{-3} \text{ T}$
 d) $4.68 \times 10^{-4} \text{ T}$
 Assume
 the part
 of the wire has the same
 charge as the
 rest of the
 wire.
 What will
 be the
 magnitude
 of the force
 exerted on
 the wire?
 a) $2.40 \times 10^{-17} \text{ N}$
 b) $1.60 \times 10^{-19} \text{ N}$
 c) $2.02 \times 10^{-14} \text{ N}$
 d) $21.0 \times 10^{-14} \text{ N}$

-21N

5.) Two charges Q_1 and Q_2 are separated at

distance of

10 mm. Calculate

the electric

field

at

distance

of

the

charges

is

Q_1 if the magnitude of Q_2 is $5 \mu\text{C}$.

a) 54×10^4

4 V b) 4.5×10^8 V c) 22×10^9 V d) 0.12×10^4

3 V

6.) A particle

of charge $+19 \mu\text{C}$ is moving

from point X to Y against

an electric

field

of

14 $\times 10^9$ N/C. Calculate

the kinetic

energy of the

particle

if

the

distance

is

1 m

and the potential difference

between X and Y is

500 V.

a) 1.33×10^6 J b) 11×10^3 J c) 2.4×10^4

J d) 13×10^6 J

7.) Assuming that the force of attraction

between

two

charges

is

ticl

esis

$1.7 \times 10^4 \text{ N}$ when they are placed 60 mm apart in space. What will

be the

magnetic

force when they are 120 mm apart

a) $11 \times 10^2 \text{ N}$ b) $17 \times 10^3 \text{ N}$ c) $6.7 \times 10^5 \text{ N}$ d) $4.$

$25 \times 10^$

5 N

8.) Suppose two parallel

wires

each carry

a current of $3.$

2 A each and are placed

4 mm apart.

Determine

the

magnitude of the magnetic force per unit

length

experienced

by the two wires.

a) $2.13 \times 10^-$

2 N/m b) $5.12 \times 10^-$

4 N/

m c) $51.$

$2 \times 10^-$

7 N/m d)

$12.5 \times 10^$

-3 N/m

9.) A wire

of length 0.5 m is

carrying a

current of 5 A and is placed

perpendicular to a

magnetic field of 0.5 T .

Calculate the

force on the

wire.

a) 2.5 N b) 1.25 N

c) 6.25 N d) 12.5 N

10.) A wire

of length 0.5 m is

carrying a

current of 5 A and is placed

perpendicular to a

magnetic field of 0.5 T .

Calculate the

force on the

wire.

a) 2.5 N b) 1.25 N

c) 6.25 N d) 12.5 N

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 a)4.8kT b)5.6kT c)7.3kT d)-2.3kT
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 a)320V/m b)225V/ m c)440V/ m d)100V/ m
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1. A 1200 μF capacitor is

connected to a

potential difference of 100 V. The

charge on the

plates is

2. A 2 mm thick plate of

aluminum is placed in a vacuum. The

area of the plate is

12. A

wire of length 1 m and cross-sectional

area 2.0 mm^2 is stretched by a

force of

100 N. The extension is

3. A particle of mass m and charge q

is moving in a

uniform

electric

field of

strength

E . The

work done by the field in moving the

particle through a distance d is

4. A

block of mass m is pushed up a

rough inclined plane of length

L and

height

h . The

work done by the

applied force is

5. A

block of

mass m is pushed up a

rough inclined plane of length

L and

height

h . The

work done

by the applied force is

6. A

block of

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a)57°C b)54°C c)55°C d)56°

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15.)Calculatethe ti

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a) 0.5m b) 3.2m c) 0.23m d) 1.1m

16.

) Alongstr

eight conductor whose cross secti

onal area is

3.3×10^{-6}

m^2 is

made of copper. It

fits a car

ries a current of 24.2 A and placed at an angl

of 45°

to the magnetic field

of $1.2 \times 10^{-3} \text{ T}$. What will be its length

h ? (assuming $g = 9.8 \text{ m/s}^2$)

he

magnitude of the

force

experienced by it is

$6.1 \times 10^{-3} \text{ N}$

3N.

a) 0.176m b) 12m c) 0.230m d) 2.3m

An electron moves at a speed of $6.2 \times 10^4 \text{ m/s}$ in a unif

orm magnetic

field

of

6.1 uT . (assuming

the motion of the

electron is

uniform)

is at a right angl

to the field

of

6.1 uT

the electron

is

17.

) The str

ength

of the magnetic

field

is

experienced by it

t

a)9.

7×10^6

-12N b)5.

52×10^6

-16N c)77.

3×10^{-6} d)6.

56×10^6

-10N

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C.

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a)3.

3×10^{-6}

10N b) 5.52×10^{-6}

16N c) 6.7×10^6

-12N d) 77.3×10^{-6}

6N

)Theear

19. t

h'sgr

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heel

ectr

on

a)0.

3×10^6

-43N b) $89.27 \times 10^{-}$

31N c) 3.

$5 \times 10^{-}$

29N d) 8.

$3 \times 10^{-}$

-30N

20.) Assuming the heel

electr

on is now moving i

ng at an angle of 33° to the magnet

ic

fi

eld. What will be the val

ue of the magneti

c for

ce it wi

ll exper

ience?

a) $55.40 \times 10^{-}$

26N b) $61.7 \times 10^{-}$

-22N c) 97.

$4 \times 10^{-}$

-26N d) 3.57×10^{-19} N

$m_e = 9.109 \times 10^{-}$

31Kg, $K_e = 8.99 \times 10^9 \text{ Nm}^2 / \text{C}^2$, $u_0 = 4\pi \times 10^{-}$

-7Tm/

A,

$g = 9.$

81 m/s^2 , $q_e = 1.$

$6 \times 10^{-}$

-19C,

$E_0 = 8.854 \times 10^{-}$

-12F/

m]

Paper 2 of 2 — phy 121 past questions

Level 100 • Second Semester • PastQuestion

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Ajagbawa Dritselaju Stephanie

GENERAL PHY 121 QUESTIONS (2017/2018)

$m_e = 9.1 \times 10^{-31} \text{ kg}$ $m_p = 1.673 \times 10^{-27} \text{ kg}$ $q_e = 1.6 \times 10^{-19}$ $E_0 = \frac{1}{36\pi} \times 10^{-9}$

- which of the following losses is caused by locally induced electromotive force?
(a) Eddy loss (b) Copper loss (c) Flux leakage (d) Hysteresis loss
- Which of the following is true about a semi-conductor?
(a) Conductivity increases with temperature
(b) Conductivity decreases with temperature
(c) Conductivity increases with pressure
(d) Conductivity is dependent on temperature and pressure
- The current flowing through the primary windings of a 200/1200 step-up transformer is 6mA. Determine the secondary side current.
(a) 0.5mA (b) 2.1mA (c) 1.0mA (d) 6mA
- A p-type semi-conductor is doped with a _____
(a) penta-valent atom (b) tri-valent atom (c) tetravalent atom (d) mono-atomic element
- The phase angle of a series R.L.C circuit is usually positive when _____
(a) The capacitive reactance is greater than the inductive reactance
(b) The inductive reactance is greater than the capacitive reactance
(c) The inductive reactance is equal to the capacitive reactance
(d) None of the above
- A $5.08 \times 10^{-12} \text{ kg}$ particle moved in an orbit perpendicular to a uniform magnetic field of 8.06 wb/m^2 . Take the charge of the particle to be $1.6 \times 10^{-19} \text{ C}$ and approximate radius of orbit as 80mm. Calculate its frequency of motion. (a) 3.4nHz (b) 4.0nHz (c) 5.3kHz (d) 4.1kHz
- An R.L.C circuit at resonance has its inductive reactance as 205.7ohm. Determine the capacitive reactance, if the resonance frequency is 62Hz.
(a) 411.4ohms (b) 102.85ohms (c) 205.7ohms (d) 12,753 ohms
- An 8.0×10^{-12} electrically charged particle moving at a speed of 50m/s is placed in a uniform magnetic field that is orthogonal to the angle of motion. Determine the magnitude of the magnetic field strength, if a force of 720N is experienced by the particle.
(a) 5.4GT (b) 11GT (c) 3.2GT (d) 1.8GT
- A conducting material whose resistance is 71.2Ω at 27°C has its resistance decreased to 20.1Ω when the temperature reaches 330k. calculate the coefficient of resistance in per kelvin.
(a) -0.127k^{-1} (b) 0.090k^{-1} (c) -0.0033k^{-1} (d) 0.08k^{-1}
- A cylindrical shape bar magnet produces a flux density of 1.12T through the cross-sectional area of the magnet as it radiates a flux of $197.3\mu\text{wb}$ all over its surface. Evaluate the diameter of the magnet. (a) 15mm (b) 1.49 m (c) 149cm (d) 0.15m
- The total number of diodes required for the construction of a bridged full wave rectifier is ____
(a) one (b) two (c) three (d) four
- Which of the following is not an application of combinational logic gate?
(a) Encoder (b) multiplexer (c) full adder (d) rectification
- Two coils have their no. of turns to be 150 & 80 each. If the cross-sectional area and length of both coils is $10.1 \times 10^{-3} \text{ m}^2$ and 170mm respectively. What is the induced mutual inductance between the two coils?
(a) 2.3mH (b) $0.9\mu\text{H}$ (c) 6.1mH (d) 1.21H
- An extrinsic semiconductor is a semiconductor in its _____ state.
(a) excitation state (b) pure state (c) doped state (d) valence state
- A covalent bonding is common to _____
(a) metals (b) metalloids (c) non-metals (d) semi-conductors
- Given the phase angle of a house hold ELEPAQ generator to be 51.2° . calculate its power factor. (a) 0.61 (b) 0.63 (c) 0.60 (d) 0.62

17. The energy possessed by a proton which experiences a magnetic force of 21.2N when moving at right angle to a magnetic field of 20.7T is
 (a) $4.1 \times 10^{-9} \text{ J}$ (b) $1.1 \times 10^{-9} \text{ J}$ (c) $15.5 \times 10^{-9} \text{ J}$ (d) $3.3 \times 10^{-9} \text{ J}$
18. If the phase angle in an electrical circuit is positive 90° this implies that the circuit is
 (a) purely capacitive (b) at resonance (c) purely inductive (d) purely resistive
19. If a proton moving at a linear speed of $2.0 \times 10^3 \text{ m/s}$ along the axis of a current carrying conductor and at a distance of 2.5cm from its center. Taking the magnitude of current to be 6.3A, calculate the magnetic field intensity B felt by the proton. (a) 2.78mT (b) 18mT (c) 0.5mT (d) 2.1mT
20. From the question given above, calculate the force that will be exerted on the proton. (a) 0.016N (b) 0.16N (c) 1.6N (d) 16N
21. Two wires A&B are placed parallel to one another carried a current of 11.12A each in the opposite direction. Determine the distance between the two wires, if the force per unit length experienced by each conductor is 12N/m. (a) $4.2 \times 10^{-6} \text{ m}$ (b) $2.1 \times 10^{-6} \text{ m}$ (c) $6.3 \times 10^{-4} \text{ m}$ (d) 5.1m
22. The force of attraction between two parallel wires carrying currents in the same direction is (a) repulsive (b) attractive (c) neutral (d) neutral
23. what will be the resulting effect when an insulating material is placed in between the parallel plates of a capacitor? (a) capacitance is transient (b) capacitance is constant (c) capacitance decreases (d) capacitance increases
24. If it takes 0.32s for a $18\mu\text{F}$ capacitor to discharge from 82pC to 54pC through a given resistor. What is the value of the resistance, assuming the time constant is 54sec?
 (a) 43kΩ (b) 43Ω (c) 21.5Ω (d) 40kΩ
25. The strength of a field force experienced by a particle placed at a distance of $4 \times 10^{-3} \text{ m}$ away from a current carrying conductor of magnitude 0.81A is
 (a) 0.041mT (b) 0.053mT (c) 0.041mT (d) 0.053mT
26. The maximum efficiency of a 11kV/220V step down domestic transformer whose high voltage side and secondary winding currents are 0.3A and 12A respectively is
 (a) 57% (b) 80% (c) 74% (d) 91%
27. What amount of work done in joules in moving a 1nC charge through two points that are 60cm apart in a magnetic field of strength $5.9 \times 10^{-3} \text{ N/C}$? (a) 0.54J (b) 0.12J (c) 0.4J (d) 0.039J
28. If an emf of 170v is induced in a coil of 18mH subjected to a varying magnetic field. Determine the rate at which current flows through the coil. (a) $9.44 \times 10^{-3} \text{ A/s}$ (b) $1.83 \times 10^{-2} \text{ A/s}$ (c) $4.71 \times 10^{-3} \text{ A/s}$ (d) $3.2 \times 10^{-2} \text{ A/s}$
29. A semiconductor has its total no. of electrons to be four in its
 (a) conduction band (b) valence band (c) energy gap (d) excitation state
30. Sufficient amount of flux linkage as much as possible in a magnetic core could be achieved with the use of
 (a) reduced no of turns (b) increased core area (c) soft iron core (d) decreased in frequency of a.c supply
31. Determine the e.m.f induced in a 142mm long rod moving at a speed of 32m/s in a magnetic field of $18 \times 10^{-2} \text{ T}$ (a) 0.82V (b) 0.45V (c) 0.67V (d) 1.12V
32. An R-L circuit has its phase angle between
 (a) 90° & 180° (b) 180° & 270° (c) 0° & 90° (d) 270° & 360°
33. A negative phase angle is usually obtained in an R-L-C circuit when the
 (a) capacitive reactance is greater than the inductive reactance (b) inductive reactance is greater than the capacitive reactance (c) capacitive reactance is less than the inductive reactance (d) resistance is equal to the circuit impedance
34. At 180° angle of rotation, the e.m.f in a coil rotating in a uniform magnetic field is
 (a) one-half (b) minimum (c) one-quarter (d) maximum
35. The conduction of current in metal is directly proportional to
 (a) strength of the conductor (b) durability of the metal (c) temperature (d) availability of free electrons
36. Find the period of motion of an electron moving in a uniform magnetic field of strength 2.25Telsa. Take the perimeter of the circular orbit as 171cm. (a) $3.4 \times 10^{-15} \text{ s}$ (b) $2.2 \times 10^{-7} \text{ s}$ (c) $1.59 \times 10^{-14} \text{ s}$ (d) $18 \times 10^{-5} \text{ s}$

37. A magnetic field of 380mT was created in a solenoid after a current of 18amps flows through its coil. Calculate the no. of turns of this coil knowing that the diameter of the solenoid coil is approximately 4.18cm . (a) 16800 turns (b) 1750 turns (c) 1840 turns (d) 1120 turns
38. Two charged particles exerted a force of 10^{11}N on each another when placed 80cm apart. At what distance should they be placed apart so as to exert a force of 10^{12}N on one another?
(a) 10mm (b) 18mm (c) 0.8mm (d) 2.34mm
39. What will be the absolute permittivity of a dielectric material whose relative permittivity is 5?
(a) 34.1F/m (b) $34.1\mu\text{F/m}$ (c) 44.2nF/m (d) 44.2pF/m
40. A $50\text{e-}50\text{mm}$ capacitor plates are separated at a distance of 0.03mm using an insulating medium whose dielectric constant is 4. Calculate the capacitance of the capacitor.
(a) 2.95nC (b) 3.0nC (c) 2.45nC (d) 1.25nC
41. In place of a magnetic field, an electric field is used instead to direct a field strength of $2.11 \times 10^{17}\text{N/C}$ on an electron. What will be the force felt by the electron within the field bounds?
(a) $3.4 \times 10^{-18}\text{N}$ (b) $1.2 \times 10^{-13}\text{N}$ (c) $2.11 \times 10^{-19}\text{N}$ (d) $3.37 \times 10^{-17}\text{N}$
42. What is turns ratio of a 33kVA transformer fed from a supply of 13MV & 1.52Amps source, if the current flowing through the external load is 60amps ? (a) 236:1 (b) 236000:1 (c) 1:236 (d) 1:236000
43. The drift current in a semiconductor is due to the _____. (a) random motion between electrons & holes (b) Thermal Effect (c) difference in applied potential (d) column pressure in the material
44. A device that can convert an Alternating current to a Direct current is known as _____.
(a) alternator (b) dynamo (c) rectifier (d) motor
45. The total power dissipated in a series R-L-C circuit at resonance is _____. (Given the following parameters: $R=50\Omega$, $L=70\text{mH}$, $C=16\mu\text{F}$ & $I_{\text{peak}}=40\text{A}$) (a) 64kW (b) 128kW (c) 96kW (d) 60kW
46. At what frequency could resonance occur in a R-L-C circuit having the following passive elements values: $R=200\Omega$, $L=15\text{mH}$ & $C=170\mu\text{F}$ (a) 50Hz (b) 100Hz (c) 150Hz (d) 200Hz
47. An external switching circuit is used to increase the rate at which current flows through a coil from 70A/s to 140A/s . The resulting effect creates an e.m.f of 205kV , find the inductance of the coil. (a) 23.4mH (b) 121mH (c) 15.43mH (d) 14.64mH
48. Which of the following transistor biasing configuration has the highest amplification factor?
(a) common base (b) common emitter (c) common collector (d) forward-reverse biasing
49. Eddy current can be described as current that flows in _____ inside a disc.
(a) pile (b) row (c) circle (d) rectangle
50. The technical term used for the voltage induced in a time varying magnetic field is known as _____.
(a) potential difference (b) e.m.f (c) alternating current (d) direct voltage
51. Electromagnetic induction is the consequence of which of the following effect?
(a) motor (b) spring (c) radial (d) thyristors
52. A negative temperature coefficient of resistance is common to which of the following?
(a) crystalloids (b) non-metal (c) metals (d) semiconductors
53. A P-N junction operating in the forward biased mode has its resistance in what order? (a) $\text{k}\Omega$ (b) $\text{M}\Omega$ (c) Ω (d) $\text{G}\Omega$
54. At ordinary temperature, an intrinsic semiconductor has _____
(a) more mobile electrons and holes (d) static electrons & holes
55. A wire made up of brass carries 1.31amps of current. At what distance away from its center will the strength of the field be $3.24\mu\text{T}$? (a) 81mm (b) 40mm (c) 34mm (d) 321mm
56. A constantan wire position in the vertical axis carries a current of 12.3A . experience a force of 3N by a magnetic field of strength 14wb/m^2 . Determine the length of the conducting wire.
(a) 4.008m (b) 0.045m (c) 0.0174m (d) 1.234m
57. A conductor of 5.4m long and cross sectional area of $2.66 \times 10^{-4}\text{m}^2$ experiences a force of $1.2 \times 10^{-4}\text{N}$ when placed in a magnetic field of $3.8 \times 10^{-6}\text{T}$. What is the angle between the conductor and the magnetic field? (assume the current flowing through the conductor is 13Amps) (a) 30.1° (b) 26.73° (c) 52.1° (d) 45.2°

37. A magnetic field of 380mT was created in a solenoid after a current of 18amps flows through its coil. Calculate the no. of turns of this coil knowing that the diameter of the solenoid coil is approximately 4.18cm. (a) 16800 turns (b) 1750 turns (c) 1840 turns (d) 1120 turns
38. Two charged particles exerted a force of $10^{11}N$ on each another when placed 80cm apart. At what distance should they be placed apart so as to exert a force of $10^{17}N$ on one another? (a) 10mm (b) 18mm (c) 0.8mm (d) 2.34mm
39. What will be the absolute permittivity of a dielectric material whose relative permittivity is 5? (a) 34.1F/m (b) 34.1 μ F/m (c) 44.2nF/m (d) 44.2 pF/m
40. A 50x50mm capacitor plates are separated at a distance of 0.03mm using an insulating medium whose dielectric constant is 4. Calculate the capacitance of the capacitor. (a) 2.95nC (b) 3.0nC (c) 2.45nC (d) 1.25nC
41. In place of a magnetic field, an electric field is used instead to direct a field strength of $2.11 \times 10^2 N/C$ on an electron. What will be the force felt by the electron within the field bounds? (a) $3.4 \times 10^{-5} N$ (b) $1.2 \times 10^{-13} N$ (c) $2.11 \times 10^{-8} N$ (d) $3.376 \times 10^{-17} N$
42. What is turns ratio of a 33kVA transformer fed from a supply of 13MV & 1.52Amps source, if the current flowing through the external load is 60amps? (a) 236:1 (b) 236000:1 (c) 1:236 (d) 1:236000
43. The drift current in a semiconductor is due to the _____. (a) random motion between electrons & holes (b) Thermal Effect (c) difference in applied potential (d) column pressure in the material
44. A device that can convert an Alternating current to a Direct current is known as _____. (a) alternator (b) dynamo (c) rectifier (d) motor
45. The total power dissipated in a series R-L-C circuit at resonance is _____. (Given the following parameters: $R=50\Omega$, $L=70mH$, $C=16\mu C$ & $I_{peak}=40A$) (a) 64kW (b) 128kW (c) 96kW (d) 60kW
46. At what frequency could resonance occur in a R-L-C circuit having the following passive elements values: $R=200\Omega$, $L=15mH$ & $C=170\mu F$ (a) 50Hz (b) 100Hz (c) 150Hz (d) 200Hz
47. An external switching circuit is used to increase the rate at which current flows through a coil from 70A/s to 140A/s. The resulting effect creates an e.m.f of 205kV, find the inductance of the coil. (a) 23.4mH (b) 121mH (c) 15.43mH (d) 14.64mH
48. Which of the following transistor biasing configuration has the highest amplification factor? (a) common base (b) common emitter (c) common collector (d) forward-reverse biasing
49. Eddy current can be described as current that flows in _____ inside a disc. (a) pile (b) row (c) circle (d) rectangle
50. The technical term used for the voltage induced in a time varying magnetic field is known as _____. (a) potential difference (b) e.m.f (c) alternating current (d) direct voltage
51. Electromagnetic induction is the consequence of which of the following effect? (a) motor (b) spring (c) radial (d) thyristors
52. A negative temperature coefficient of resistance is common to which of the following? (a) crystalloids (b) non-metal (c) metals (d) semiconductors
53. A P-N junction operating in the forward biased mode has its resistance in what order? (a) k Ω (b) M Ω (c) Ω (d) G Ω
54. At ordinary temperature, an intrinsic semiconductor has _____. (a) more holes (b) more electrons (c) few mobile electrons and holes (d) static electrons & holes
55. A wire made up of brass carries 1.31 amps of current. At what distance away from its center will the strength of the field be 3.24 μ T. (a) 81mm (b) 40mm (c) 34mm (d) 321mm
56. A constantan wire positioned in the vertical axis carries a current of 12.3A experience a force of 3kN by a magnetic field of strength 14Twb/m². Determine the length of the conducting wire. (a) 4.008m (b) 0.045m (c) 0.0174m (d) 1.234m
57. A conductor of 5.4m long and cross sectional area of $2.66 \times 10^{-4} m^2$ experiences a force of $1.2 \times 10^{-4} N$ when placed in a magnetic field of $3.8 \times 10^{-6} T$. What is the angle between the conductor and the magnetic field? (assume the current flowing through the conductor is 13Amps) (a) 30.1 $^\circ$ (b) 26.73 $^\circ$ (c) 52.1 $^\circ$ (d) 45.2 $^\circ$

58. A solenoid connected to an 17sinwt A.C source dissipated 213J of heat energy. What is the resistance of the solenoid? (a) 20.23Ω (b) 43.23Ω (c) 17.72Ω (d) 11.1Ω
59. If an electron moving with a speed of 32m/s overlap into a magnetic field of 16.24Telsa in magnitude. What will be the force exerted on the electron? (Take the angle of deviation to be 26.4°) (a) $3.7 \times 10^{-17}\text{N}$ (b) $4.2 \times 10^{-12}\text{N}$ (c) $11 \times 10^{-6}\text{N}$ (d) $9.0 \times 10^{-10}\text{N}$
60. A $4.5\mu\text{F}$ ceiling fan capacitor discharges through the coils of its stator windings. If the resistance of the stator winding is calculated to be 142Ω from the equivalent circuit. Determine the time constant. (a) 0.45m/s (b) 0.91m/s (c) 0.52m/s (d) 0.64m/s
61. The flux density produced from the core of a coil of 1600 turn copper wire when the current of 42A flows through its winding is _____. (a) 8.1mT (b) 8.4mT (c) 4.25mT (d) 5.72mT
62. The conductivity of metal decreases with _____. (a) increases with temperature (b) decreases with temperature (c) remain constant (d) pulsate with temperature
63. Conduction takes place in semiconductors due to the presence of _____. (a) electrons (b) electrons and holes (c) holes (d) none of the above
64. A charge of 39C flows through an aluminum wire of cross-sectional area of 11.4mm^2 in 18secs. Calculate the drift velocity V_d . (Take the no. of free electrons to be 1.21×10^{20}) (a) 9.8km/s (b) 1000m/s (c) 9600m/s (d) 9.7km/s
65. _____ is the opposition to any change in flow of current through a coil. (a) admittance (b) resistivity (c) inductance (d) impedance
66. Evaluate the root mean square value of an e.m.f induced in a coil, if its peak-peak voltage is 464V (a) 82V (b) 164V (c) 212V (d) 328V
67. A current induced in a coil placed in a time varying magnetic field depends on the following except? (a) speed of motion (b) coil resistance (c) flux density (d) core size
68. The capacitance of an electrolytic capacitor is calculated to be $20\mu\text{C}$. Determine the amount of energy stored by the capacitor when a potential difference of 15v is applied across its terminal. (a) 13.6mJ (b) 4.12mJ (c) 2.25mJ (d) 3.21mJ
69. Which of the following is true about the force of attraction between two electrically charged particles separated in space? (a) it increases with increase in distance (b) it decreases with increase in magnitude of charge (c) it increases with decreases in distance (d) it increases with decreases in magnitude of charge
70. _____ is a device that acts like a switch (a) diode (b) resistor (c) inductor (d) transistor

(2018/2019) GENERAL PHY 121 SOLUTIONS

ANSWERS

1. A
2. A
3. Given;
 $N_p = 200$ $N_s = 1200$ $I_p = 6 \text{ mA}$
 Recall: For an ideal transformer

$$\frac{N_p}{N_s} = \frac{I_s}{I_p} \quad ; \quad \frac{200}{1200} = \frac{I_s}{6 \text{ mA}}$$

$$I_s = \frac{200 \times 6}{1200} = 1 \text{ mA} \quad (C)$$
4. B Tri-Valent Impurities
5. B
6. Given;
 $M = 5.08 \times 10^{-11} \text{ kg}$ $B = 8.06 \text{ T}$
 $q = 1.6 \times 10^{-19} \text{ C}$ $r = 80 \text{ mm} = 0.08 \text{ m}$
 Knowing that:
 The perimeter of the circular orbit "d"
 $d = 2\pi r$
 $\therefore r = \frac{d}{2\pi}$ Also: Period $T = \frac{d}{v}$ from $v = \frac{d}{T}$
 So that $f = \frac{1}{T} = \frac{v}{d}$ — (A)

6. Recall;

$$F_{\text{magnetic}} = qvB = F_{\text{centripetal}} = \frac{mv^2}{r}$$

$$qvB = \frac{mv^2}{\frac{d}{2\pi}}$$

$$qvB = \frac{2\pi mv^2}{d}$$

$$f = \frac{v}{d} = \frac{qB}{2\pi m}$$

$$f = \frac{1.6 \times 10^{-19} \times 8.06}{2 \times 3.142 \times 5.08 \times 10^{-11}}$$

$$= 4.04 \times 10^{-9} \text{ Hz} \quad (B)$$

7. At resonance

$$X_L = X_C$$

$$X_C = 205.7 \Omega$$

OPTION

(C)

8. Given;

$$Q = 8 \times 10^{-12} \text{ C}$$

$$V = 50 \text{ m/s}$$

$$F = qvB$$

$$F_{\text{magnetic}} = 720 \text{ N}$$

$$B = ?$$

8

$$B = \frac{F}{qv}$$

$$B = \frac{720}{8 \times 10^{-12} \times 50} = 1.8 \times 10^{12} \text{ T}$$

$$B = 1.8 \text{ GT}_{\text{class}} \quad (b)$$

9. Given

$$R_1 = 71.2 \Omega \quad \theta_1 = 27^\circ \text{C} = 300 \text{ K}$$

$$R_2 = 20.1 \Omega \quad \theta_2 = 330 \text{ K}$$

$$\alpha = ?$$

Keen:

$$\frac{R_1}{R_2} = \frac{1 + \alpha \theta_1}{1 + \alpha \theta_2}$$

$$\frac{71.2}{20.1} = \frac{1 + \alpha(300)}{1 + \alpha(330)}$$

$$3.54(1 + 330\alpha) = 1 + 300\alpha$$

$$3.54 + 1168.2\alpha = 1 + 300\alpha$$

$$762.69\alpha = -2.54$$

$$\alpha = -0.0033 \text{ K}^{-1} \quad (c)$$

10. Given:

$$B = 1.12 \text{ T}$$

$$\phi = 197.3 \times 10^{-6} \text{ wb}$$

Recall:

$$B = \frac{\phi}{A}$$

$$A = \frac{\phi}{B} = \frac{197.3 \times 10^{-6}}{1.12}$$

$$A = 1.762 \times 10^{-4} \text{ m}^2$$

for a cylinder Cross Section Area 

$$A_{\text{area}} = \pi \frac{D^2}{4}$$

$$\text{Diameter} = \sqrt{\frac{4 A_{\text{area}}}{\pi}} = \sqrt{\frac{4 \times 1.762 \times 10^{-4}}{3.142}} = 5.61 \times 10^{-5}$$

$$D = \underline{\underline{0.0149 \text{ m}}} \quad (\text{A})$$

11. 12. 

13. Given;

$$N_1 = 150$$

$$N_2 = 180$$

$$A = 10.1 \times 10^{-2} \text{ m}^2$$

$$l = 170 \text{ mm} = 0.17 \text{ m}$$

4. Given

$$F_m = 23.2$$

$$B = 20.7 \text{ T}$$

$$q_p = 1.6 \times 10^{-19}$$

$$E = 7$$

$$m_p = 1.673 \times 10^{-27} \text{ kg}$$

$$E = \frac{1}{2} m v^2$$

$$\text{from } F = q_p B = \frac{m v^2}{r}$$

$$v = \frac{F}{q_p B}$$

$$\text{§ } M = m_{\text{proton}}$$

$$v = \frac{23.2}{1.6 \times 10^{-19} \times 20.7} = 7 \times 10^{18} \text{ m/s}$$

$$E = \frac{1}{2} \times 1.673 \times 10^{-27} \times (7 \times 10^{18})^2$$

$$E = 4.1 \times 10^{10} \text{ Joules}$$

(A)

13. Recall: $N_0 = 4\pi \times 10^{-7} \text{ A/m}$

$$M = \frac{N_0 N_1 N_2 A}{L}$$

$$= \frac{4\pi \times 10^{-7} \times 150 \times 80 \times 1.01 \times 10^{-3}}{0.14 \text{ m}}$$

$$M = 8.96 \times 10^{-5} \text{ H}$$

$$M = \underline{\underline{0.9 \mu\text{H}}} \quad (B)$$

14. C

15. D

16. Given $\phi = 51.2$

$$P.F. = \cos \phi = \cos 51.2 = \underline{\underline{0.627}} \approx \underline{\underline{0.63}} \quad (B)$$

18.

C

19

$$V_p = 2 \times 10^{23} \text{ m/s} \quad d = 2.5 \text{ cm} = 0.025 \text{ m}$$

$$I = 6.3 \text{ Amps} \quad B = ?$$

$$\text{Where } N_0 = 4\pi \times 10^{-7} \text{ A/m}$$

$$B = \frac{N_0 I}{2\pi d} = \frac{4\pi \times 10^{-7} \times 6.3}{2\pi \times 0.025}$$

$$B = 0.504 \times 10^{-6} \text{ T}$$

$$= 0.504 \mu\text{T} \quad (C)$$

20. From Question 19.

$$V_{\text{proton}} = 2.0 \times 10^{23} \text{ m/s} \quad q_{\text{proton}} = 1.6 \times 10^{-19} \text{ C}$$

$$B = 0.5 \times 10^{-6} \text{ T}$$

$$F_{\text{EXERTED}} = q_{\text{proton}} \times V_{\text{proton}} \times B$$

$$= 1.6 \times 10^{-19} \times 2.0 \times 10^{23} \times 0.5 \times 10^{-6}$$

$$F_{\text{EXERTED}} = \underline{0.016 \text{ N}} \quad (A)$$

21. Given: $I_A = I_B = 11.12 \text{ Amps}$
 $\frac{F}{L} = 12 \text{ N/m}$
 distance apart (r) = ?

Recall: $\frac{F}{L} = \frac{\mu_0 I_A I_B}{2\pi r}$

$$12 = \frac{4\pi \times 10^{-7} (11.12)^2}{2\pi (r)}$$

$$r = \frac{10^{-7} (11.12)^2}{6}$$

$$r = 2.1 \times 10^{-6} \text{ m}$$

 (B.)

22. B ATTRACTIVE

23. D CAPACITANCE INCREASES

24. Given:

$$t = 0.32 \text{ s}$$

$$C = 18 \times 10^{-6} \text{ F}$$

$$R = ?$$

$$Q_1 = 82 \times 10^{-6} \text{ C}$$

$$Q_2 = 54 \times 10^{-6} \text{ C}$$

$$L = 54 \text{ sec} = R$$

$$Q_2 = Q_1 e^{-t/RC}$$

$$\ln \frac{Q_2}{Q_1} = -\frac{t}{RC}$$

$$-\frac{C}{t} \left[\ln \frac{Q_2}{Q_1} \right] = -\frac{1}{R}$$

$$-\frac{18 \times 10^{-6}}{0.32} \left[\ln \frac{54 \times 10^{-6}}{82 \times 10^{-6}} \right] = -\frac{1}{R}$$

$$R = \frac{1}{2.35 \times 10^{-5}}$$

$$R \approx \underline{\underline{43 \text{ k}\Omega}}$$

(A)

23) Given:

$$d = 4 \times 10^{-3} \text{ m}$$

$$I = 0.81 \text{ Amp}$$

$$B = \frac{N I}{2\pi d}$$

$$B = \frac{4\pi \times 10^{-7} \times 0.81}{2\pi \times 4 \times 10^{-3}}$$

$$B = 0.41 \times 10^{-4} \text{ T}$$

$$= \underline{0.041 \text{ mT}} \quad (C)$$

26) $V_p = 11,000 \text{ V}$

$V_s = 220 \text{ V}$

$I_p = \frac{550}{0.3 \text{ A}}$

$I_s = 12 \text{ Amp}$

$$\% = \frac{\text{Output Power}}{\text{Input Power}} = \frac{V_s \times I_s}{V_p \times I_p}$$

$$\text{Transformer Efficiency} = \frac{220 \times 12}{11,000 \times 0.3} \times 100\%$$

$$= 80\% \quad (B)$$

27.

$$Q = 11 \times 10^{-9} \text{ C}$$

$$d = 60 \times 10^{-2} \text{ m}$$

$$E = 5.9 \times 10^6$$

Recall:

$$\text{Work Done} = Q E d$$

$$= 11 \times 10^{-9} \times 5.9 \times 10^6 \times 60 \times 10^{-2}$$

$$= 38941 \times 10^{-5}$$

$$= \underline{\underline{0.039 \text{ J}}} \quad (\text{D})$$

28.

$$E.m.f = 170 \text{ V}$$

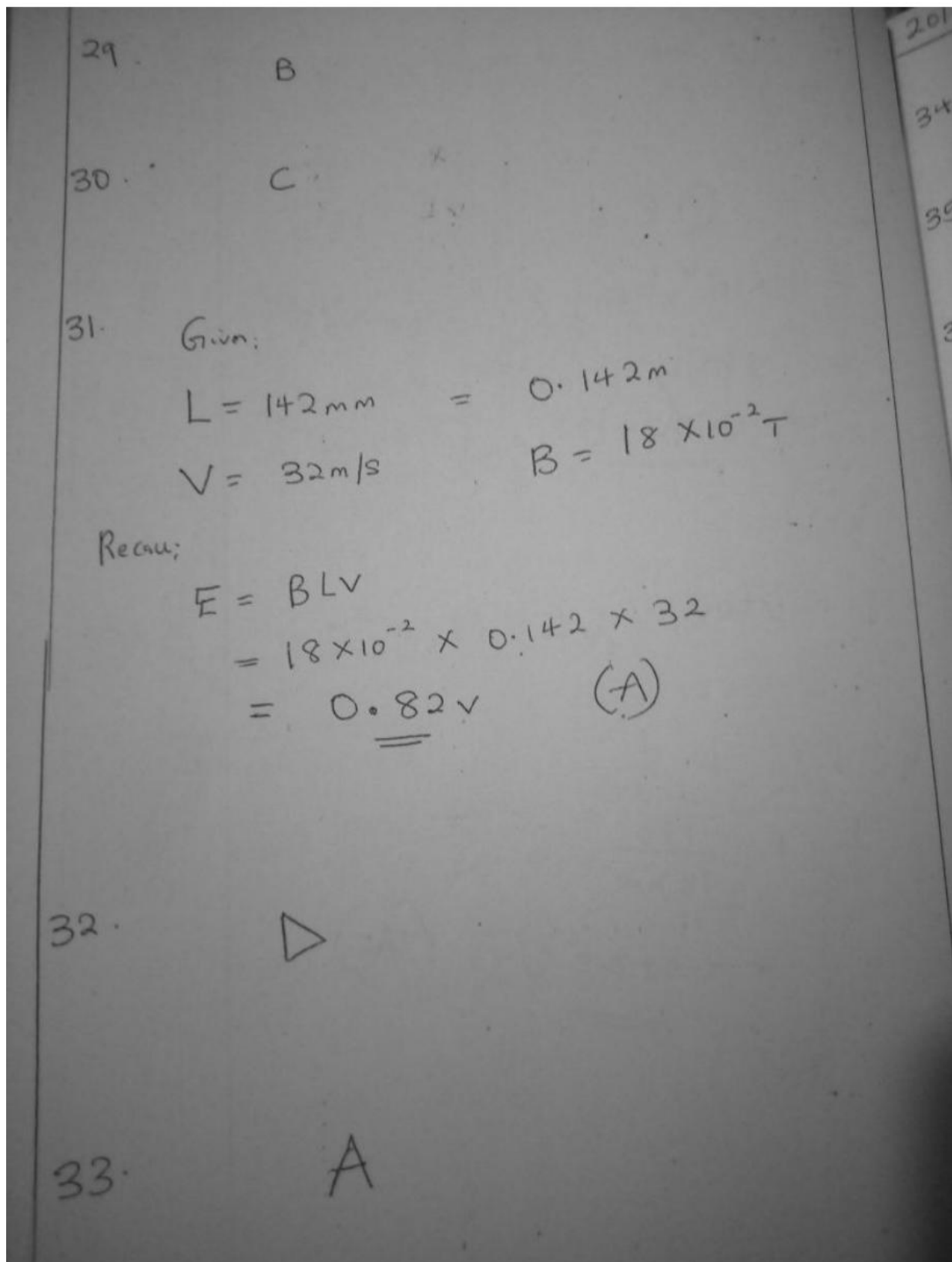
$$L = 18 \times 10^{-3} \text{ H}$$

Recall:

$$E.m.f = L \frac{di}{dt}$$

$$\frac{di}{dt} = \frac{E}{L} = \frac{170}{18 \times 10^{-3}}$$

$$= \underline{\underline{9.44 \times 10^3 \text{ A/s}}} \quad (\text{A})$$



2017/2018 GENERAL PHY 121 SOLUTIONS

34. B

35. Δ

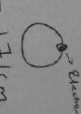
36. Given:

$B = 2.25 \text{ T}$

$q_e = 1.6 \times 10^{-19}$

$m_e = 9.1 \times 10^{-31} \text{ kg}$

$d = 2\pi r = 17 \text{ cm} = 0.17 \text{ m}$

r_{center} 

$F = qvB = \frac{mv^2}{r}$

$qB = \frac{mv(2\pi)}{2\pi r}$

$qB = \frac{m(2\pi)v}{d}$

$T = \frac{d}{v} = \frac{2\pi m_e}{qB}$

$= \frac{2 \times 3.142 \times 9.1 \times 10^{-31}}{1.6 \times 10^{-19} \times 2.25}$

$= 1.59 \times 10^{-11} \text{ s.}$

(c)

37.

Given

$$B = 380 \text{ mT} = 380 \times 10^{-3} \text{ T}$$

$$I = 18 \text{ amps}$$

$$d = 4.18 \text{ cm} = 0.0418 \text{ m}$$

$$N = ?$$

Note:

$$B = \mu_0 N I$$

$$N = \frac{B}{\mu_0 I} = \frac{380 \times 10^{-3}}{4\pi \times 10^{-7} \times 18}$$

$$N = \underline{\underline{16800 \text{ turns}}}$$

(A)

38.

Given:

$$F_1 = 10^{11} \text{ N}$$

$$d_1 = 800 \text{ cm} = 0.8 \text{ m}$$

$$F_2 = 10^{17} \text{ N}$$

$$d_2 = ?$$

Provided k, q_1 & q_2 are constant

$$F_1 \propto \frac{1}{d_1^2} \quad \text{so that} \quad F_1 d_1^2 = F_2 d_2^2$$

$$\therefore d_2^2 = \frac{F_1 d_1^2}{F_2} = \frac{10^{11} \times (0.8)^2}{10^{17}} = 10^{-6} \times 0.64$$

$$d_2^2 = 640 \times 10^{-3} \text{ mm}$$

$$d_2 = \sqrt{640 \times 10^{-3}}$$

$$d_2 = \underline{\underline{0.8 \text{ mm}}}$$

(C)

39. Given

$$\epsilon_r = 5 \quad \epsilon_0 = 8.84 \times 10^{-12} \text{ F/m}$$

Absolute Permittivity: $\epsilon = \epsilon_0 \epsilon_r$

$$\epsilon = 5 \times 8.84 \times 10^{-12}$$

$$\epsilon = 44.21 \text{ pF/m}$$

(D)

40. Given:

$$\begin{aligned} \text{Area (A)} &= 50 \times 50 \text{ mm}^2 \\ &= 2500 \text{ mm}^2 = 2500 \times 10^{-6} \text{ m}^2 \\ &= 2.5 \times 10^{-3} \text{ m}^2 \end{aligned}$$

$$\begin{aligned} \text{Distance (d)} &= 0.03 \text{ mm} \\ &= 0.03 \times 10^{-3} \text{ m} \\ &= 3 \times 10^{-5} \text{ m} \end{aligned}$$

$$\epsilon_r = 4$$

$$\epsilon_0 = 8.84 \times 10^{-12} \text{ F/m}$$

$$\epsilon = \epsilon_r \epsilon_0 = 35.36 \times 10^{-12} \text{ F/m}$$

Recall:

$$C = \frac{A \epsilon}{d} = \frac{2.5 \times 10^{-3} \times 35.36 \times 10^{-12}}{3 \times 10^{-5}}$$

$$C = 29.47 \times 10^{-10} \text{ C}$$

$$C = 2.95 \text{ nC}$$

(A)

41. $E = 2.11 \times 10^3 \text{ N/C}$ $q_e = 1.6 \times 10^{-19} \text{ C}$
 $F = q_e \times E = 1.6 \times 10^{-19} \times 2.11 \times 10^3$
 $F_{\text{Electric}} = 3.376 \times 10^{-17} \text{ N}$ (A)

42. $V_p = 13 \times 10^6 \text{ V}$ $i_p = 3 \text{ Amps}$
 $P_s = \text{Output Power} = 18 V_s = 33 \times 10^3 \text{ VA}$
 $i_s = 600 \text{ Amps}$
 $V_s = \frac{P_s}{i_s} = \frac{33 \times 10^3}{600} = 55 \text{ V}$
 Regu;
 Turns Ratio = $\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{13 \times 10^6}{55}$
 $= 236,000 : 1$ (B)

43. C

44 C

45 Note that at Resonance

$$X_L = X_C$$

$$\text{Impedance } Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$\boxed{Z = R}$$

Given

$$i_{\text{max}} = 40 \text{ A} \quad \text{and} \quad R = 80 \Omega$$

When

$$i_{\text{rms}} = \frac{i_{\text{peak}}}{\sqrt{2}} = \frac{40}{\sqrt{2}} = 28.284$$

(A)

$$\text{Power} = i_{\text{rms}}^2 R$$

$$= 28.284^2 \times 80 = 64 \text{ kW}$$

46.

Given;

$$L = 15 \text{ mH} = 15 \times 10^{-3} \text{ H}$$

$$C = 170 \text{ nF} = 170 \times 10^{-6} \text{ F}$$

$$f_{\text{resonance}} = \frac{1}{2\pi\sqrt{LC}} = \frac{1}{2\pi\sqrt{15 \times 10^{-3} \times 170 \times 10^{-6}}}$$

$$\underline{f = 99.65 \text{ Hz}} \quad \text{(B)}$$

47.

Given:

$$\frac{di}{dt} = 14 \times 10^{-3} \text{ A/s}$$

$$E = 205 \times 10^3 \text{ V}$$

When

$$E = L \frac{di}{dt}$$

$$L = \frac{E}{\frac{di}{dt}} = \frac{205 \times 10^3}{14 \times 10^{-3}}$$

(D)

$$L = 14642 \text{ H}$$

$$14.64 \text{ MH}$$

48.

B

49

C

50

B

51.

A

52.

53

54.

55

52. Δ

53. C

54. C

$$55. \quad I = 1.31 \text{ A}$$

$$N_0 = 4\pi \times 10^{-7}$$

$$B = 3.24 \times 10^{-6} \text{ T}$$

$$r = ?$$

$$B = \frac{N_0 I}{2\pi r}$$

$$r = \frac{N_0 I}{2\pi B} = \frac{4\pi \times 10^{-7} \times 1.31}{2\pi \times 3.24 \times 10^{-6}}$$

$$r = 0.081 \text{ m}$$

$$= \underline{\underline{81 \text{ mm}}}$$

(A)

56.

$$\vec{i} = 12.3 \hat{i}$$

$$\vec{F} = 3 \hat{k} \text{ N}$$

$$L = ?$$

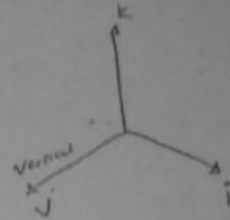
$$\vec{B} = -14 \hat{i} \text{ T/m}$$

Recall:

$$\vec{F} = i \vec{L} \times \vec{B}$$

$$3 \hat{k} = 12.3 (\vec{L} \times -14 \hat{i})$$

$$\vec{L} = \frac{3 \hat{k}}{12.3 \cdot (-14 \hat{i})} = 0.0174 \text{ m } \hat{j} \quad (C)$$



57. Given

$$L = 5.4 \text{ m}$$

$$A_{\text{area}} = 2.66 \times 10^{-4} \text{ m}^2$$

$$\vec{F} = 1.2 \times 10^{-4} \text{ N}$$

$$\vec{B} = 3.8 \times 10^{-6} \text{ T}$$

$$I = 13 \text{ Amps}$$

$$F = BIL \sin \theta$$

$$\sin \theta = \frac{F}{BIL} = \frac{1.2 \times 10^{-4}}{3.8 \times 10^{-6} \times 13 \times 5.4}$$

$$\theta = \sin^{-1}(0.4498)$$

$$\theta = 26.73^\circ$$

(B)

58.

$$i_{\text{peak}} = 17 \text{ A}$$

$$E = 213 \text{ J}$$

$$E = i_{\text{rms}}^2 \cdot R$$

$$R = \frac{E}{i_{\text{rms}}^2} = \frac{213}{17^2} \cdot \sqrt{2} = 17.72 \Omega$$

(C)

59.

Given

$$V = 32 \text{ m/s}$$

$$B = 16.24 \text{ T}$$

$$q_e = 1.6 \times 10^{-19} \text{ C}$$

$$\theta = 26.4^\circ$$

$$F = qVB \sin \theta$$

$$= 1.6 \times 10^{-19} \times 32 \times 16.24 \times \sin 26.4^\circ$$

$$F = 3.697 \times 10^{-17} \text{ N}$$

= (A)

60. Given $C = 4.5 \times 10^{-6} \text{ F}$ $R = 142 \Omega$
 $\tau = ?$

Recall; the Time Constant of an R-C Circuit

$$\tau = RC$$

$$\tau = 4.5 \times 10^{-6} \times 142 \Omega$$

$$= 639 \times 10^{-6} \text{ s}$$

$$\tau = 0.64 \text{ ms}$$

(D)

61. Given

$$N = 160 \quad I = 42 \text{ A}$$

$$\mu_0 = 4\pi \times 10^{-7} \text{ Tm/A}$$

$$B = \mu_0 \times N \times I$$

$$= 4\pi \times 10^{-7} \times 160 \times 42$$

$$B = 8.4 \text{ mT}$$

(B)

62. B.

63. B

$$64. \quad Q = 39 \text{ C} \quad A = 11.4 \text{ mm}^2$$

$$= 11.4 \times 10^{-6} \text{ m}^2$$

$$t = 18 \text{ s}$$

$$V_d = 7 \quad n = 1.21 \times 10^{20}$$

Recall;

$$Q = I t$$

$$I = \frac{Q}{t} = \frac{39}{18} = 2.17 \text{ Amps}$$

Where

$$I = n A V_d e$$

$$V_d = \frac{I}{n A e}$$

$$V_d = \frac{2.17}{1.21 \times 10^{20} \times 11.4 \times 10^{-6} \times 1.6 \times 10^{-19}}$$

$$V_d = 0.0983 \times 10^5$$

$$V_d = \underline{\underline{9832.2 \text{ m/s}}}$$

65. C

$$66 \quad I_{\text{peak-peak}} = 464$$

$$I_{\text{peak}} = \frac{I_{\text{peak-to-peak}}}{2}$$

$$I_{\text{r.m.s}} = \frac{I_{\text{peak}}}{\sqrt{2}} = \frac{I_{\text{peak-to-peak}}}{2\sqrt{2}}$$

$$V_{\text{r.m.s}} = \frac{464}{2\sqrt{2}} = 164 \text{ V} \quad (\text{B})$$

67. $\triangleright$

68. $C = 20 \mu\text{C} = 20 \times 10^{-6} \text{C}$

$$V = 15 \text{V}$$

$$E_{\text{energy}} = \frac{1}{2} C V^2$$

$$= \frac{1}{2} \times 20 \times 10^{-6} \times 15^2$$

$$= \underline{\underline{2.25 \text{ mJ}}} \quad (\text{C})$$

69. C

70. A \$ \triangleright

GENERAL PHY 121 QUESTIONS

- The conductivity of a metal depends on which of the following particles? (a) Neutron (b) Electrons (c) positron (d) deuterium
- A perfect diode acts like a (a) switch (b) Resistor (c) capacitor (d) amplifier
- A 100 turn rectangular coil has a dimension of 200mm by 5mm. If it is placed at right angle to a uniform magnetic field whose flux density changes from $19\mu\text{T}$ to 0.2mT within 16s, calculate the e.m.f induced in the coil within the given period of time. (a) 2.1V (b) 1.3V (c) 0.12V (d) 12.0V
- An e.m.f induced in a coil is minimal when the magnetic flux is at degree to the plane of coil area. (a) 180 (b) 90 (c) 450 (d) 270
- An alternating current whose peak value is 1.32A flows through a 120 Ω resistor, determine the r.m.s value of the voltage drop across it. (a) 112V (b) 231V (c) 122 V (d) 416V
- A given circular coil of resistance 2.5 Ω and area 300cm^2 is placed in a uniform magnetic field of strength 2.5mT in such a way that the plane will receive maximum flux. Assuming the plane of coil is now rotated through an angle of 90°, calculate the change in flow of charge through it. (a) 40 μC (b) 30 μC (c) 50 μC (d) 60 μC
- Which of the following transistor configuration is the most commonly used in amplification circuit. (a) common collector (b) common base (c) common emitter (d) none of the above
- The phase angle of an R-L-C circuit is when the inductive reactance is greater than the capacitive reactance. (a) positive (b) negative (c) zero (d) 360°
- Two charges Q1 and Q2 are separated in space through a distance of 10mm. Calculate the electric field intensity experienced by charge Q2 due to charge Q1 if the magnitude of Q2 is 5 μC . (a) $4.5 \times 10^8\text{V}$ (b) $22 \times 10^8\text{V}$ (c) $54 \times 10^8\text{V}$ (d) $6.12 \times 10^8\text{V}$
- A given coil has an inductance of 2.5H and the flow of current through it changes from 5mA to 2.1mA in 1 μs . Calculate the back e.m.f induced in the coil. (a) 4.25kV (b) 3.25kV (c) 1.24kV (d) 7.25kV
- Which of the following circuit has a phase angle between 90° and 0°? (a) R-C (b) R-L-C (c) R-L (d) all of the above.
- A soft magnetic material is used to prevent which of the following transformer losses. (a) copper loss (b) hysteresis loss (c) eddy current loss (d) amperian loss
- Calculate the maximum amount of current flowing through a 5mH inductor, if it stores 50 joules of energy. (a) 134A (b) 141A (c) 125A (d) 116A
- The phase angle in a purely inductive circuit is (a) -90° (b) +90° (c) 0° (d) -180°
- A 20 μF capacitor has a capacitive reactance of 132.6 Ω when connected to an a.c source. Determine the frequency of the a.c source? (a) 50Hz (b) 60Hz (c) 120 Hz (d) 100Hz
- Determine the velocity at which a given rod of length 2.5cm should move in a magnetic field $\vec{B} \times 10^{-5}\text{T}$ for an e.m.f of $6 \times 10^{-4}\text{V}$ to be induced in it. (a) 110m/s (b) 220m/s (c) 400m/s (d) 350m/s
- What is the phase angle of an R-L-C circuit whose power factor is 0.36? (a) 69° (b) 71° (c) 70° (d) 23°
- The current flowing through a wire rotating in a magnetic field does not depend on which of the following. (a) angular velocity of the coil (b) resistance of the coil (c) magnetic flux density (d) none of the above.
- The average power dissipated in a series R-L-C circuit where $R=50\Omega$, $L=10\text{mH}$, $C=15\mu\text{F}$ and $I_{\text{rms}}=10\text{A}$ is (a) 5kJ (b) 15kJ (c) 20kJ (d) 11kJ
- In the construction of a transformer a soft iron core is preferably used to allow the maximum passage of (a) Current (b) e.m.f (c) magnetic flux (d) heat loss.
- Which of the following type of bond exist in a semiconductor atom? (a) coordinate covalent bond (b) covalent bond (c) electrovalent bond (d) all of the above
- Determine the r.m.s value of the current flowing through a 50 Ω resistor and a 20mH inductor connected in series to a 220V, 50Hz a.c supply. (a) 1.11A (b) 2.32A (c) 4.36A (d) 55A
- The loss in a transformer due to locally induced e.m.f is known as (a) copper loss (b) hysteresis loss (c) eddy current loss (d) flux leakage
- At room temperature the maximum number of electrons in the valence band of a semiconductor is (a) 3 (b) 4 (c) 4 (d) 2
- A device that generates an alternating current from a mechanical energy input is called a (a) alternator (b) dynamo (c) inductor (d) motor
- A semiconductor in its pure state is usually referred to as (a) extrinsic conductor (b) intrinsic semiconductor (c) a p-type material (d) a neutral particle
- The properties of a coil in which it opposes any change in flow of current through it is known as (a) resistance (b) inductance (c) reluctance (d) permittivity

38. In metals, conduction of current is due to the movement of electrons. While in semiconductors, conduction is due to the movement of (a) holes (b) mobile electrons (c) only energized electrons (d) free electrons and holes
39. The rate of change in the current flowing through a 20mH inductor is 50(A/sec). Determine the value of the induced EMF in the coil. (a) 100V (b) 10V (c) 1000V (d) 1V
40. The diffusion process which involves the random motion of holes and free electrons in semiconductors is known as (a) external pressure (b) thermal agitation (c) potential difference applied (d) all of the above
41. The length and area of two similar coils A and B placed near each other are 0.25m and $1.4 \times 10^{-3} \text{ m}^2$ respectively. Determine the mutual inductance between the coils, if the number of turns of coil A and B are 15 and 15 respectively. (a) $5.3 \times 10^{-4} \text{ H}$ (b) $5.3 \times 10^{-3} \text{ H}$ (c) $5.3 \times 10^{-2} \text{ H}$ (d) $5.3 \times 10^{-1} \text{ H}$
42. At room temperature, a pure silicon semiconductor acts like (a) an insulator (b) a metal (c) a semiconductor (d) a ferrite
43. A transformer has a turn ratio of 16:1 and a current of 4Amps is flowing through its secondary winding. Determine the primary side current. (a) 0.88A (b) 1.88A (c) 4.2A (d) 9.0A
44. A particle of charge $+19\mu\text{C}$ is moving from a point X to Y against a uniform electric field of 1400 V/m . Calculate the kinetic energy of the particles when it returns to point Y. Let the distance between X and Y be 500cm. (a) $2.4 \times 10^3 \text{ J}$ (b) $1.3 \times 10^3 \text{ J}$ (c) $1.1 \times 10^3 \text{ J}$ (d) $1.3 \times 10^2 \text{ J}$
45. An ideal conductor was wound on an insulating material such that it has 2000 turns and a current of 1.17A flows through it. What is the magnitude of the magnetic field it generates? (a) 0.64T (b) 3.3T (c) 0.84T (d) 1.1T
46. Assuming the force of attraction between two charged particles is $1.7 \times 10^4 \text{ N}$ when they are placed 60mm apart in space. What will be the magnitude of force when they are 120mm apart? (a) $1.1 \times 10^4 \text{ N}$ (b) $1.7 \times 10^4 \text{ N}$ (c) $6.7 \times 10^4 \text{ N}$ (d) $4.2 \times 10^4 \text{ N}$
47. Suppose two parallel wires carrying a current of 3.2Amps each are placed 4mm apart. Determine the magnitude of the magnetic force per unit length experienced by the two wires. (a) 0 (b) 0.3 (c) 0.4 (d) 0.5
48. The force in Question 47 is (a) attractive (b) repulsive (c) neutral (d) none of the above
49. An aluminum wire is carrying a current of 1.6A. Determine the strength of the magnetic field at a point 0.07m away from its center. (a) 3.3mT (b) 1.2mT (c) 4.5mT (d) 2.9mT
50. The capacitance of a capacitor tends to be when its plates are separated. (a) an insulating material compared to when they are placed in vacuum (b) larger (c) smaller (d) none of the above
51. The electric field intensity between the parallel plates of a capacitor is $16 \times 10^3 \text{ V/m}$ in vacuum. What will be the magnitude of the electric field intensity when a dielectric material of thickness 6 mm is placed between the plates? Assume the dielectric constant to be 5. (a) 520V/m (b) 440V/m (c) 100V/m (d) 25V/m
52. A wire of length 0.5m lying along the x-axis carries a steady current of 2.5A and it experiences a magnetic force of -6N at a right angle to its axis. Determine the strength of the magnetic field intensity. (a) 4.8kT (b) 5.6kT (c) 7.3kT (d) 2.3kT
53. Estimate the area of the each of the plates of a capacitor that is required to build a $1200\mu\text{F}$ capacitor. If the plates are to be separated at a distance of 4.2mm in vacuum. (a) $43\pi \times 10^{-3} \text{ m}^2$ (b) $570\pi \times 10^{-3} \text{ m}^2$ (c) $670\pi \times 10^{-3} \text{ m}^2$ (d) $90\pi \times 10^{-3} \text{ m}^2$
54. A long straight conductor whose cross sectional area is $3.3 \times 10^{-6} \text{ m}^2$ is made of copper. If it carries a current of 24.2A and placed at an angle of 45° to the magnetic field of $1.2 \times 10^{-2} \text{ T}$. What will be its length? Assume the magnitude of the force experienced by it is $3.61 \times 10^{-3} \text{ N}$. (a) 0.230m (b) 0.176m (c) 12m (d) 2.3m
55. Calculate the time it will take a capacitor of capacitance $30\mu\text{F}$ to discharge from an initial value of $120\mu\text{C}$ through a $5\text{k}\Omega$ resistor for a time constant of 50secs. (a) 0.25m (b) 1.1m (c) 3.2m (d) 0.5m
56. If the resistance of a solenoid is 3.92 and a current of 9.37A flows through it. What is the amount of energy dissipated in form of heat? (a) 3331J (b) 342J (c) 243J (d) 176J
57. Determine the amount of charge that will pass through an aluminum wire of cross sectional area 2.0mm^2 in 6 hours, if the electrons drift velocity is $1.62 \times 10^{-4} \text{ m/s}$. Take the number of free electrons to be $2.1 \times 10^{23} \text{ m}^{-3}$. (a) 54C (b) 66C (c) 71C (d) 54C
58. An electron travelling at a velocity of 4.7 m/s enters a magnetic field B whose magnitude is 0.06131 T . Take what will be the magnitude of the force on the electron? (a) $4.3 \times 10^{-13} \text{ N}$ (b) $5.7 \times 10^{-12} \text{ N}$ (c) $6.7 \times 10^{-14} \text{ N}$ (d) $1.98 \times 10^{-17} \text{ N}$
59. A coil has a resistance of 62.5Ω at 7°C , at what temperature will its resistance increase to 71.5Ω . Take the coefficient of resistance of the material to be 0.017°C^{-1} . (a) 54°C (b) 59°C (c) 56°C (d) 57°C
60. The magnitude of the magnetic field force experienced by a particle placed at a distance of $3 \times 10^{-3} \text{ m}$ away from a long straight conductor which carries a current of 0.72 Amp is (a) 0.048T (b) 0.039T (c) 2.1T (d) 100T

- An electron moves at a speed of $6.3 \times 10^6 \text{ m/s}$ in a uniform magnetic field of $66 \mu\text{T}$. Assuming the motion of the electron is at right angle to the field. Determine: (question 81-86)
81. The strength of the magnetic force experienced by it. (a) $77.3 \times 10^{-9} \text{ N}$ (b) $6.56 \times 10^{-19} \text{ N}$ (c) $9.7 \times 10^{-12} \text{ N}$ (d) $5.52 \times 10^{-16} \text{ N}$
 82. Assume the magnetic field is replaced by an electric field of 3450 N/C . What will be the force on the electron? (a) $977.3 \times 10^{-6} \text{ N}$ (b) $3.3 \times 10^{-19} \text{ N}$ (c) $6.7 \times 10^{-12} \text{ N}$ (d) $5.52 \times 10^{-16} \text{ N}$
 83. The earth gravitational pull on the electron. (a) $0.3 \times 10^{-43} \text{ N}$ (b) $3.5 \times 10^{-29} \text{ N}$ (c) $8.3 \times 10^{-39} \text{ N}$ (d) $89.27 \times 10^{-31} \text{ N}$
 84. Which of the forces has the largest magnitude?
 85. Which of the forces has the least magnitude?
 86. Assuming the electron is now moving at an angle of 33° to the magnetic field. What will be the value of the magnetic force it will experience? (a) $97.4 \times 10^{-26} \text{ N}$ (b) $3.57 \times 10^{-19} \text{ N}$ (c) $61.7 \times 10^{-22} \text{ N}$ (d) $55.40 \times 10^{-28} \text{ N}$
 57. A $40 \mu\text{F}$ capacitor was discharged through a $180 \text{ k}\Omega$ resistor, estimate the time constant. (a) 5.1 (b) 6.5 (c) 7.2 (d) 5.6
 58. The magnetic flux generated by a 200 turns per meter wire when a current of 3.08 A flows through it is: (a) 0.77 mT (b) 0.88 mT (c) 1.1 mT (d) 0.66 mT
 59. A rectangular shaped electromagnet is to produce a flux density of 4.8 T through its surface. If it generates a flux of 921.5 mWb , what will be the dimension of the electromagnet? (let the length to breadth ratio be 4:3) (a) $0.21 \text{ m} \times 0.063 \text{ m}$ (b) $0.11 \text{ m} \times 0.083$ (c) $0.5 \text{ m} \times 0.3 \text{ m}$ (d) $0.76 \text{ m} \times 0.2 \text{ m}$
- A particle of mass $6.27 \times 10^{-28} \text{ kg}$ is moving in a circular orbit at right angle to a uniform magnetic field of 10.2 T and its radius of orbit is 90 cm . Calculate: (question 60-63)
60. The period of motion about the orbit. (a) $3.51 \times 10^{-13} \text{ s}$ (b) $4.57 \times 10^{-12} \text{ s}$ (c) $11.7 \times 10^{-22} \text{ s}$ (d) $4.4 \times 10^{-15} \text{ s}$
 61. The amount of energy possessed by the particle. (a) $17.1 \times 10^{13} \text{ eV}$ (b) $3.07 \times 10^9 \text{ eV}$ (c) $9.7 \times 10^{10} \text{ eV}$ (d) $5.1 \times 10^{11} \text{ eV}$
 62. The frequency of the orbital motion. (a) $1.12 \times 10^{14} \text{ Hz}$ (b) $0.28 \times 10^{13} \text{ Hz}$ (c) $5.7 \times 10^{12} \text{ Hz}$ (d) $3.2 \times 10^{20} \text{ Hz}$
 63. A particle of charge $6.1 \mu\text{C}$ and mass $5.4 \times 10^{-18} \text{ kg}$ is subjected to a uniform magnetic field B directed to its velocity. Calculate the magnitude of the magnetic field required to accelerate the particle through a velocity of 200 m/s . (Assume the radius of the circular orbit in which it moves is 4.1 cm) (a) $4.31 \times 10^{-7} \text{ T}$ (b) $3.5 \times 10^{-1} \text{ T}$ (c) $2.7 \times 10^{-1} \text{ T}$ (d) $0.1 \times 10^{-7} \text{ T}$
- Chapter 4**
- A electrically charged particle is moving with a velocity of $2.7 \times 10^8 \text{ m/s}$ along the axis of a conductor carrying a current of 11 A . If the particle is at a distance of 4.7 mm from the center of the wire. Determine: (question 64-69)
64. The magnitude of the magnetic field intensity experienced by the particle. (a) $7.4 \times 10^{-16} \text{ T}$ (b) $4.2 \times 10^8 \text{ T}$ (c) $4.58 \times 10^{-4} \text{ T}$ (d) $2.31 \times 10^{-6} \text{ T}$
 65. Assuming the particle has the same charge as an electron, what will be the magnitude of force it experiences? (a) $2.40 \times 10^{-17} \text{ N}$ (b) $1.60 \times 10^{-19} \text{ N}$ (c) $21.0 \times 10^{-21} \text{ N}$ (d) $20.22 \times 10^{-19} \text{ N}$
 66. What amount of current should flow through a solenoid of 330 turns and diameter 4.5 cm for it to produce a magnetic field strength of 400 mT ? (a) 965 A (b) 762 A (c) 324 A (d) 198 A
 67. A wire of length 160 cm is moving at a velocity of 16 m/s in a magnetic field of $3 \times 10^{-12} \text{ Tesla}$. Calculate the magnitude of the emf induced in the coil. (a) $7.17 \times 10^{-11} \text{ V}$ (b) $7.68 \times 10^{-18} \text{ V}$ (c) $11.7 \times 10^{-13} \text{ V}$ (d) $2.41 \times 10^{-6} \text{ V}$
 68. A $1500 \text{ W}/220 \text{ V}$ transformer has an efficiency of 89% . Determine the amount of current flowing through its primary winding. Where $I_p = 50 \text{ mA}$. (a) 45.5 mA (b) 53.1 mA (c) 82.4 mA (d) 71.5 mA
 69. The applications of combinational logic gate include which of the following? (a) Encoder (b) multiplexers (c) full adders (d) all of the above.
 70. A diode made of germanium is used in a full wave rectifier circuit to supply a resistive load of 220Ω from an ac supply of 230 V . Calculate the peak diode current. (a) 1.1 A (b) 3.2 A (c) 2.6 A (d) 5.2 A

GENERAL PHY 121 SOLUTIONS

ANSWERS

OPTION (B)

OPTION (A)

SOLUTION

Given Length = 200 mm & Breadth = 5 mm
 $\therefore \checkmark \text{ Area} = (200 \times 5) \text{ mm}^2 = 1000 \text{ mm}^2$

Also, $\phi_1 = 19 \text{ mT}$ and $\phi_2 = 0.2 \text{ mT}$

$$\checkmark d\phi = (19 - 0.2) \text{ mT} = 18.8 \text{ mT}$$

$$dt = 16 \text{ s}$$

$$\text{cell; } E_{\text{m.f}} = N \frac{d\phi}{dt} = 100 \left(\frac{18.8 \times 10^{-3}}{16} \right)$$

$$E = \underline{0.12 \text{ V}} \quad (C)$$

(A)

Given; $I_{\text{max}} = 1.32 \text{ A}$ & $R = 120 \Omega$

$$I_{\text{r.m.s}} = \frac{I_{\text{max}}}{\sqrt{2}} = \frac{(0.7071 \times 1.32) \text{ A}}{\sqrt{2}} = 0.933 \text{ A}$$

$$V_{\text{r.m.s}} = I_{\text{r.m.s}} \times R = 1.20 \times 0.933$$

$$V_{\text{r.m.s}} = \underline{112.01 \text{ V}} \quad (A)$$

⑥ Given: $R = 2.5 \Omega$, $B = 2.5 \text{ mT}$
 Recall: $E = \frac{N d\psi}{dt} = \frac{NBA}{dt}$ where $N = 1$
 $E = \frac{BA}{dt} = IR = \frac{d\phi}{dt} \cdot R$
 $d\phi = \frac{BA}{R}$ Given Area = 300 cm^2
 $d\phi = \frac{2.5 \times 10^{-3} \times 300 \times 10^{-4}}{2.5}$
 $d\phi = \underline{\underline{30 \mu\text{C}}} \quad (\text{B})$

⑦ (C) Common EMITTER

⑧ OPTION (A) POSITIVE

⑨ Given: $r = 10 \text{ mm} = 10^{-2} \text{ m}$
 $Q_2 = 5 \times 10^{-6} \text{ C}$
 Recall: $E = \frac{Q_2}{4\pi\epsilon_0 r^2} = \frac{5 \times 10^{-6}}{4\pi \times \frac{1}{36\pi} \times 10^{-9} \times (10^{-2})^2}$
 $E = \frac{5 \times 10^{-6}}{\frac{r}{9} \times 10^{-9} \times 10^{-4}}$
 $E = \underline{\underline{45 \times 10^7 \text{ V}}} \quad (\text{A})$

⑩ Given $L = 2.5 \text{ H}$ $\frac{dI}{dt} = 10^{-6} \text{ s}$ $\frac{dI}{dt} = (5-2.1) \text{ mA}$
 Recall, $E = L \frac{dI}{dt} = \frac{2.5 (2.9 \times 10^{-3})}{10^{-6}}$
 $= 7.25 \text{ kV (B)}$

⑪ Option (C) R-L

⑫ (B) Hysteresis loss

⑬ Given: $L = 5 \text{ mH}$ or $5 \times 10^{-3} \text{ H}$
 $E = 50 \text{ J}$
 Recall: $E_L = \frac{1}{2} L I_{\text{max}}^2$
 $I_{\text{max}} = \sqrt{\frac{2E}{L}} = \sqrt{\frac{2 \times 50}{5 \times 10^{-3}}}$
 $I_{\text{max}} = 141.42 \text{ A (B)}$

⑭ Option (B) $+90^\circ$

⑮ Given $C = 20 \times 10^{-6} \text{ F}$ $X_C = 132.6 \Omega$
 Recall: $X_C = \frac{1}{2\pi fC}$

$f = \frac{1}{2\pi \times 10^{-4}} = \frac{1}{2 \times 3.14 \times 10^{-4} \times 20 \times 10^{-4}}$
 $f = \underline{60 \text{ Hz}} \quad (B)$

16) Given: $L = 0.025 \text{ m}$ $B = 6 \times 10^{-5} \text{ T}$
 $E = 6 \times 10^{-4} \text{ V}$
 Recall: $E = BLv \therefore v = \frac{E}{BL}$

$v = \frac{6 \times 10^{-4}}{6 \times 10^{-5} \times 0.025} = 400 \text{ m/s} \quad (C)$

17) Recall: $\cos \phi = 0.86$
 $\phi = \cos^{-1} 0.86$
 $= \underline{68.89^\circ}$

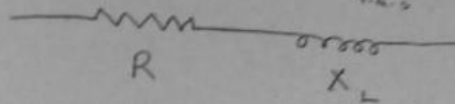
18) (D)

19) Given $R = 50 \Omega$ $I_{\text{rms}} = 10 \text{ A}$
 Note: Power is dissipated only in the resistor
 Recall: $P_{\text{av}} = I_{\text{rms}}^2 R$
 $P_{\text{av}} = 10^2 \times 50 = 5 \times 10^3 \text{ J}$
 $= \underline{5 \text{ kJ}} \quad (A)$

20 Magnetic flux (C)

21 B

22 Given; $R = 50 \Omega$ $L = 20 \times 10^{-3} \text{ H}$
 $f = 50 \text{ Hz}$ $V_{\text{r.m.s}} = 220$



Recall; $X_L = 2\pi fL = 2 \times 3.142 \times 50 \times 20 \times 10^{-3}$
 $= 6.284 \Omega$

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{50^2 + 6.284^2}$$

$$= 50.39 \Omega$$

$$I_{\text{r.m.s}} = \frac{V_{\text{r.m.s}}}{Z} = \frac{220}{50.39} = 4.36 \text{ A} \quad (\text{C})$$

23 (C)

24 (C)

25 (A)

26 (B)

27. B

28. D

29. Given; $\frac{di}{dt} = 500 \text{ A/s}$ & $L = 20 \text{ mH}$

Recall;

$$E_{\text{m.f.}} = L \frac{di}{dt} = 20 \times 10^{-3} \times 500$$

$$E = \underline{\underline{10 \text{ V}}} \quad (\text{B})$$

30. C

31. Given $A = 1.4 \times 10^{-2} \text{ m}^2$
 $L = 0.25 \text{ m}$
 $N_1 = 50$ & $N_2 = 15$
 $\mu_0 = 4\pi \times 10^{-7} \text{ A/m}$

Recall;

$$M = \frac{\mu_0 N_1 N_2 A}{L}$$

$$M = \frac{4\pi \times 10^{-7} \times 50 \times 15 \times 1.4 \times 10^{-2}}{0.25}$$

$$M = \underline{\underline{52.79 \times 10^{-6} \text{ H}}} \quad (\text{B})$$

32. C

33. Given

$$N_p = 16, N_s = 1$$

$$I_s = 14 \text{ A} \quad \& \quad I_p = ?$$

Recall;

$$\frac{N_p}{N_s} = \frac{I_s}{I_p} \quad ; \quad \frac{16}{1} = \frac{14}{I_p}$$

$$I_p = \frac{14 \times 1}{16} = 0.875 \text{ A} \quad (\text{A})$$

34. Given

$$Q = +19 \mu\text{C}, E = 14 \times 10^9$$

$$\text{and } S = 0.5 \text{ m}$$

Work done (W) in moving from point X to Y is equal to the kinetic energy (E_k) of the particle at Point Y.

$$W = E_k = QES = 19 \times 10^{-6} \times 14 \times 10^9 \times 0.5$$

$$= 1.33 \times 10^5 \text{ J} \quad (\text{B})$$

35. Given

$$N = 2000$$

$$I = 333$$

$$B = ?$$

Recall;

$$B = \mu_0 I N$$

$$B = 4\pi \times 10^{-7} \times 333 \times 2000$$

$$B = 0.838 \text{ Telsa} \quad (\text{C})$$

36. Given $F_1 = 1.7 \times 10^6 \text{ N}$ $r_1 = 60 \times 10^{-3} \text{ m}$
 $F_2 = ?$ $r_2 = 120 \times 10^{-3} \text{ m}$

Recall; $F_1 \propto \frac{1}{r_1^2}$ $\therefore F_1 r_1^2 = K = F_2 r_2^2$

$$F_2 = \frac{F_1 r_1^2}{r_2^2} = \frac{1.7 \times 10^6 \times (0.06)^2}{(0.12)^2}$$

$$F_2 = \underline{\underline{0.425 \times 10^6 \text{ N}}} \quad (\text{D})$$

37. Given $I_1 = I_2 = 3.2 \text{ A}$
 $r = 4 \times 10^{-3} \text{ m}$

Recall; $\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi r}$

$$= \frac{2 \times 10^{-7} \times (3.2)^2}{4 \times 10^{-3}}$$

$$\frac{F}{L} = 5.12 \times 10^{-4} \text{ N}$$

38. ATTRACTIVE (A).

39. Given; $I = 1.6 \text{ A}$ $r = 0.07 \text{ m}$
 Recall; $B = \frac{\mu_0 I}{2\pi r} = \frac{4\pi \times 10^{-7} \times 1.6}{2\pi \times 0.07}$
 $B = \underline{4.57 \mu\text{T}} \quad (\text{C})$

40. A

41. Given $E_0 = 16 \times 10^2 \text{ V/m}$
 $\epsilon_r = 5$

Recall; $E = \left(\frac{Q}{A}\right) \frac{1}{\epsilon}$ Where $\frac{Q}{A} = \frac{\text{Charge}}{\text{Area}}$

is Constant.

$$E_0 \propto \frac{1}{\epsilon_0}; \quad E_0 \epsilon_0 = E_1 \epsilon_0 \epsilon_r \quad \text{--- (1)}$$

E_0 = Electric field intensity in Vacuum

E_1 = " " " " in dielectric material

ϵ_0 = Permittivity of free space

$$E_1 = \frac{E_0}{\epsilon_r} \quad \text{--- (2) from eqn (1)}$$

42. $E_1 = \frac{16 \times 10^2}{5} = 3.2 \times 10^2 \text{ V/m}$ (A)

43. Given: $C = 1200 \text{ nF}$
 $d = 4.2 \times 10^{-3} \text{ m}$
 Area = ? $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$

Recall;

$$C = \frac{A \epsilon_0}{d}$$

$$A = \frac{C d}{\epsilon_0} = \frac{1200 \times 10^{-6} \times 4.2 \times 10^{-3}}{8.84 \times 10^{-12}}$$

$$= 570.14 \times 10^3 \text{ m}^2 \text{ (B)}$$

42. Given $L = 0.5 \text{ m}$
 $I = 2.5 \text{ A}$
 $B = ?$ $F = -6 \hat{j} \text{ N}$

Recall;

$$\vec{F} = I \vec{B} \times \vec{L}$$

$$\vec{F} = I \vec{L} \times \vec{B}$$

42. $-6\hat{j} = 2.5(0.5\hat{i} \times \vec{B})$
 $-2.4\hat{j} = 0.5\hat{i} \times \vec{B}$
 Recall: $\hat{i} \times \hat{k} = -\hat{j}$
 $\therefore \vec{B} = \frac{-2.4\hat{j}}{0.5\hat{i}} = \underline{\underline{4.8\hat{k} \text{ T (A)}}}$

44. Given:
 $A = 3.3 \times 10^{-6} \text{ m}^2$
 $I = 24.2 \text{ A}$
 $L = ?$
 $\theta = 45^\circ$
 $B = 1.2 \times 10^{-3} \text{ T}$
 $F = 3.61 \times 10^{-3} \text{ N}$

Recall: $F = BIL \sin \theta$
 $L = \frac{F}{BI \sin \theta} = \frac{3.61 \times 10^{-3}}{1.2 \times 10^{-3} \times 24.2 \times \sin 45^\circ}$

$L = \underline{\underline{0.176 \text{ m (B)}}}$

45. Given; $R = 5000 \Omega$
 $Q_0 = 120 \times 10^{-6} \text{ C}$ $Q_1 = 23 \times 10^{-6} \text{ C}$
 $C = 30 \times 10^{-6} \text{ F}$ $\tau = 5000 \times 30 \times 10^{-6}$
 or RC

Recall;

$$Q_1 = Q_0 e^{-t/RC}$$

$$23/120 = 120 e^{-t/0.15}$$

$$0.192 \text{ } \cancel{5.22} = e^{-t/0.15}$$

$$\ln \frac{0.192}{\cancel{5.22}} = -t/0.15$$

$$-1.652 = -t/0.15$$

$$t = 0.248 \text{ s} \quad (A)$$

46. Given $R = 3.9 \Omega$

$$I = 9.37 \text{ A}$$

$$E_{\text{energy}} = I^2 R$$

46. $E = 9.37^2 \times 3.9$
 $= 342.41 \text{ Joules (B)}$

47. Given: $A = 2 \times 10^{-6} \text{ m}^2$, $t = 6.1 \text{ s}$
 $V_d = 1.62 \times 10^4 \text{ m/s}$ $n = 2.1 \times 10^{21}$

Recall $I = n A V_d e$

$I = 2.1 \times 10^{21} \times 2 \times 10^{-6} \times 1.62 \times 10^4 \times 1.6 \times 10^{-19}$
 $= 10.89 \text{ A}$

Also;

$Q = I t$

$Q = 10.89 \times 6.1$

$Q = 66.43 \text{ C}$
 $=$

(C)

48. Given $V = 4 \hat{i} \text{ m/s}$
 $B = (0.06 \hat{i} - 31 \hat{k}) \text{ T}$

48. $q = 1.6 \times 10^{-19}$

$$F = q (v \times B)$$

$$F = 1.6 \times 10^{-19} (4\hat{i} \times (0.06\hat{i} - 31\hat{j}))$$

$$F = 1.6 \times 10^{-19} (-124(-\hat{j}))$$

$$= \underline{\underline{1.98 \times 10^{-17} \hat{j} \text{ N}}} \quad (D)$$

49. Given: $R_1 = 62.5 \Omega$, $\theta_1 = 7^\circ \text{C}$
 $= 280 \text{ K}$

$R_2 = 71.5 \Omega$ & $\theta_2 = ?$

$\alpha = 0.017 \text{ K}^{-1}$

Recall;

$$\frac{R_1}{R_2} = \frac{1 + \alpha \theta_1}{1 + \alpha \theta_2}$$

$$\frac{62.5}{71.5} = \frac{1 + 0.017(280)}{1 + 0.017(\theta_2)}$$

$$0.874 = \frac{5.76}{1 + 0.017\theta_2}$$

49.

$$0.874 + 0.0149 \theta_2 = 5.76$$

$$0.0149 \theta_2 = 4.89$$

$$\theta_2 = 328.19 \text{ K} \quad (B)$$

$$\theta_2 = 328.19 - 273 = \underline{\underline{55.19^\circ \text{C}}}$$

$$50. \quad r = 3 \times 10^{-6} \text{ m}$$

$$I = 0.72 \text{ A}$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{4\pi \times 10^{-7} \times 0.72}{2\pi \times 3 \times 10^{-6}}$$

$$= \underline{\underline{0.48 \times 10^{-1} \text{ T}}} \quad (A)$$

51.

Given;

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$v = 6.2 \times 10^4 \text{ m/s}$$

$$B = 66.1 \mu\text{T}$$

$$F_m = q v B$$

$$= 1.6 \times 10^{-19} \times 6.2 \times 10^4 \times 66.1 \times 10^{-6}$$

$$= 6.56 \times 10^{-19} \text{ N (B)}$$

==

52.

$$F_E = q E$$

$$= 1.6 \times 10^{-19} \times 3450$$

$$= 5.52 \times 10^{-16} \text{ N}$$

==

(D)

53.

Recall

$$m_e = 9.1 \times 10^{-31} \text{ kg}$$

$$g = 9.81 \text{ m/s}^2$$

$$F_g = m_e \times g.$$

$$= 9.1 \times 10^{-31} \times 9.81$$

$$(D) = 89.27 \times 10^{-31} \text{ N}$$

54. Electric field force

55. Gravitation force

$$56. F_m = q v B \sin \theta$$

$$= (6.56 \times 10^{-19}) \sin 33^\circ$$

$$= 3.57 \times 10^{-19} \text{ N}$$

$$= (B)$$

57. Given $C = 40 \text{ nF}$
 $R = 180 \times 10^3 \Omega$

$$\tau = RC$$

$$= 40 \times 10^{-6} \times 180 \times 10^3$$

$$= \underline{\underline{7.2 \text{ (s)}}}$$

58. Given $N = 200$
 $I = 3.08 \text{ A}$

$$B = \mu_0 N I$$

$$= 4\pi \times 10^{-7} \times 200 \times 3.08$$

$$= \underline{\underline{0.77 \text{ mT (A)}}}$$

59. Given $B = 4.8 \text{ T}$
 $\phi = 0.924 \text{ wb}$

59.

Recall;

$$B = \frac{\phi}{A}$$

$$A = \frac{\phi}{B} = \frac{0.924}{4.8}$$

$$A = 0.193 \text{ m}^2$$

Also; $L : B = 4 : 3$

$$\text{Length} = \frac{4}{7} \times 0.193 = 0.11 \text{ m}$$

$$\text{Breadth} = \frac{3}{7} \times 0.193 = 0.083 \text{ m}$$

$$\therefore \text{Dimension} \Rightarrow 0.11 \text{ m} \times 0.083 \text{ m} = (B)$$

60. Given; $m = 6.27 \times 10^{-28} \text{ kg}$

$$B = 10.2 \text{ T}$$

$$r = 0.9 \text{ m}$$

let the distance travelled around the orbit
once be $d = ?$

60. $d = 2\pi r$ (Circular Orbit)
 And $v =$ Speed of the particle
 $q = 1.1 \times 10^{-15} \text{ C}$

Recall;

$$F = qvB = \frac{mv^2}{r} \quad \text{--- (i)}$$

Magnetic force = Centripetal force
 [from the field] $\Leftrightarrow$ [used in accelerating round the orbit]

From eqn (i)

$$v^2 = \frac{qBr}{m}$$

$$v = \frac{qBr}{m}$$

$$= \frac{1.1 \times 10^{-15} \times 10.2 \times 0.9}{6.27 \times 10^{-28}}$$

$$v = 1.61 \times 10^{13} \text{ m/s}$$

$$d = 2 \times 3.142 \times 0.9 = 5.66 \text{ m}$$

$$v = \frac{d}{T}$$

$$T = \frac{d}{v}$$

$$T = \frac{5.66}{1.61 \times 10^{13}}$$

$$T = 3.51 \times 10^{-13} \text{ s} \quad (A)$$

61.

$$K.E = \frac{1}{2} m v^2$$

$$= \frac{1}{2} \times 6.27 \times 10^{-28} \times (1.61 \times 10^{13})^2$$

$$E_k = 8.13 \times 10^{-13} \text{ J}$$

$$= 5.1 \times 10^{17} \text{ eV} \quad (D)$$

62.

$$f = \frac{1}{T} = \frac{1}{3.51 \times 10^{-13}}$$

$$= 0.28 \times 10^{13} \text{ Hz} \quad (B)$$

63.

Given;

$$Q = 6.1 \times 10^{-6} \text{ C}$$

$$m = 5.4 \times 10^{-18} \text{ kg}$$

$$v = 200 \text{ m/s}$$

$$r = 0.041 \text{ m}$$

Recall;

$$qvB = \frac{mv^2}{r}$$

$$6.1 \times 10^{-6} \times \cancel{200} \times B = \frac{(200)^2 \times 5.4 \times 10^{-18}}{0.041}$$

$$B = \frac{200 \times 5.4 \times 10^{-18}}{6.1 \times 10^{-6} \times 0.041}$$

$$B = 4.31 \times 10^{-9} \text{ T (A)}$$

64.

Given

$$v = 2.7 \times 10^9 \text{ m/s}$$

$$I = 11 \text{ A}$$

$$d = 4.7 \times 10^{-3} \text{ m}$$

Recall: $B = \mu_0 N I$

$$I = \frac{B}{\mu_0 N}$$

$$= \frac{400 \times 10^{-3}}{4\pi \times 10^{-7} \times 330}$$

$$I = 964.45 \text{ A} \quad (A)$$

67. Given:

$$L = 0.16 \text{ m}, \quad V = 16 \text{ m/s}$$

$$B = 3 \times 10^{-12} \text{ T}$$

$$\theta = 66^\circ$$

$$F = B L \sin \theta$$

66 Recall: $B = \mu_0 N I$

$$I = \frac{B}{\mu_0 N}$$

$$= \frac{4\pi \times 10^{-7} \times 330}{4\pi \times 10^{-7} \times 330}$$

$$I = 964.45 \text{ A}$$

(A)

67 Given:

$$L = 0.16 \text{ m}, V = 16 \text{ m/s}$$

$$B = 3 \times 10^{-12} \text{ T}$$

$$\theta = 66^\circ$$

$$F = B I L \sin \theta$$

167. $E = B \cdot v$
 $E = 3 \times 10^{-12} \times 0.16 \times 16$
 $= 7.68 \times 10^{-12} \text{ V} \quad (B)$

68 $V_p = 1500 \text{ V}$
 $V_s = 220 \text{ V}$
 $\eta = 89\%$
 $I_s = 500 \text{ mA}$

$$\eta = \frac{\text{Output Power}}{\text{Input Power}} = \frac{V_s \cdot I_s}{V_p \cdot I_p}$$

$$\eta = 0.89 = \frac{0.5 \times 220}{1500 \times I_p}$$

$$I_p = \frac{110}{1500 \times 0.89}$$

68 $82.4 \text{ mA} = I_p (C)$

69. Δ

70. Given $R = 220 \Omega$
 $V = 230 \sin(300t) \text{ V}$

$$V_m = 230 \text{ V}$$

$$I_m = \frac{V_m}{R} = \frac{230}{220}$$

$$= 1.045 \text{ A (A)}$$

